Name: Shobha Gupta

Case Study: Communicating Epistemic Uncertainty in the Iraq Family Health Survey (2008)

Based on: "Violence-Related Mortality in Iraq from 2002 to 2006." NEJM 358.5 (2008): 484-493.

Introduction

This case study uses the framework outlined by van der Bles et al. (2019) to evaluate how uncertainty was conveyed in the 2008 Iraq Family Health Survey (IFHS) published in *The New England Journal of Medicine*. The IFHS aimed to estimate the number of deaths due to violence in Iraq following the U.S.-led invasion in March 2003. The central estimate was approximately 151,000 violent deaths, but the authors provided a wide uncertainty range—from 104,000 to 223,000—based on statistical modeling and adjustments for incomplete data. This makes the study an ideal example of communicating *epistemic uncertainty*, which refers to uncertainty due to incomplete knowledge.

1. Who Communicated the Uncertainty?

The uncertainty was communicated by the **Iraq Family Health Survey Study Group**, which included public health officials from Iraq's Ministry of Health and experts from the **World Health Organization (WHO)**. These were domain experts who had direct access to the data and were responsible for designing the survey, analysing the results, and presenting the findings.

In van der Bles et al.'s framework, these communicators are categorized as the *"owners of the uncertainty"*. Importantly, they made a conscious effort to stay neutral and factual, especially given the political sensitivity of estimating wartime deaths. Their role was not to make policy recommendations but to provide scientifically grounded information while being transparent about the study's limitations.

2. What Was Communicated?

Object of Uncertainty

The focus of the uncertainty communication was a **numerical estimate**: the number of violence-related deaths in Iraq from March 2003 to June 2006.

Sources of Uncertainty

Several challenges contributed to the uncertainty:

- **Underreporting**: Some households may not have reported all deaths due to fear, migration, or complete dissolution of the household after a death.
- **Missing Data**: 10.6% of household clusters were not surveyed due to security concerns, especially in high-risk regions like Anbar and Baghdad.
- **Outdated Population Data**: The sampling relied on projections from older censuses, which were potentially inaccurate due to large-scale migration.
- **Recall Bias**: Respondents were asked to recall deaths that occurred over a period of years, leading to possible errors or omissions.

These are classic examples of **epistemic uncertainty** arising from data quality limitations and gaps in knowledge, as described by van der Bles et al.

Name: Shobha Gupta

Level and Magnitude of Uncertainty

The uncertainty was communicated both **directly** and **indirectly**:

- **Direct**: Through statistical expressions such as confidence intervals and uncertainty ranges. For example, the estimated violence-related death rate was 1.67 per 1,000 person-years, with a 95% uncertainty range.
- **Indirect**: By discussing the limitations of the data collection process and comparing results with other studies.

The **magnitude** of uncertainty was substantial: the central estimate was 151,000 violent deaths, but the full uncertainty range stretched from **104,000 to 223,000** deaths. This was derived using **Monte Carlo simulations** that accounted for multiple factors like survey sampling error, underreporting, and migration.

3. In What Form Was the Uncertainty Communicated?

Expression Types

The study used several types of uncertainty expressions:

- **Type ii**: Summaries of probability distributions (e.g., confidence intervals for death rates).
- **Type iii**: Rounded or order-of-magnitude estimates (e.g., "151,000 deaths").
- **Type vii**: Expressions of humility and acknowledgment of limitations (e.g., "All methods presented here have shortcomings").

Format

- **Numerical**: Statistical ranges and confidence intervals were used to show the variability in the estimates.
- **Verbal**: Phrases like "likely underreported," "recall bias may contribute," and "assumes a stable population" were used to describe the uncertainty in more accessible terms.
- **Visual**: Graphs (e.g., Figure 1B in the article) compared different estimates over time and by source, visually showing how widely the estimates varied.

These align well with van der Bles et al.'s recommended practices for clear and effective uncertainty communication, including transparency, numerical grounding, and multiple formats for diverse audiences.

4. To Whom Was the Uncertainty Communicated?

The communication targeted multiple audiences:

- **Academic and Medical Communities**: Published in *The New England Journal of Medicine*, the primary readers included epidemiologists, health policy experts, and public health researchers.
- **Policy Makers**: Including government officials, international agencies, and humanitarian organizations who needed reliable data for decision-making.
- **The Informed Public and Media**: The findings were widely reported and used to inform global discussions about the war's human cost.

Given the political context, different audiences could interpret the findings differently. Some might see the uncertainty as a sign of caution and credibility, while others might interpret it as doubt or weakness in the findings.

5. To What Effect?

Cognitive Impact (Understanding)

The communication helped readers understand not just the estimated number of deaths, but also how confident the researchers were in their estimates. It promoted *numeracy*, allowing the audience to grasp the limits of what could be known.

Emotional Impact

The large range and the high numbers likely triggered emotional responses such as shock, sadness, or concern. However, the balanced tone of the report helped prevent panic or sensationalism.

Impact on Trust

By openly discussing the study's limitations, the researchers built **credibility**. Compared to the **Burnham et al. (2006)** study (which estimated 601,000 deaths), the IFHS appeared more measured and scientifically cautious. The use of transparent methods and acknowledgment of uncertainty likely increased **trust** in the findings.

Policy Impact

The IFHS estimates were considered more reliable by international agencies like the **United Nations** and **World Health Organization**, influencing how the humanitarian crisis in Iraq was understood and addressed.

This outcome reflects van der Bles et al.'s finding that *acknowledging uncertainty does not necessarily reduce public trust*—and can even enhance it when done transparently and responsibly.

Conclusion

The Iraq Family Health Survey is a strong example of responsible and effective communication of epistemic uncertainty. The authors:

- Clearly explained what they did and didn't know
- Used numerical ranges, visual comparisons, and verbal qualifiers
- Compared their findings with other studies
- Avoided overconfidence or political bias

Using the van der Bles et al. framework shows the power of clear, honest uncertainty communication—especially in complex, high-stakes situations like wartime mortality. The IFHS demonstrates what the authors refer to as **"muscular uncertainty"**: presenting rigorous, careful estimates without hiding the unknowns. This approach supports informed decision-making while maintaining public trust.